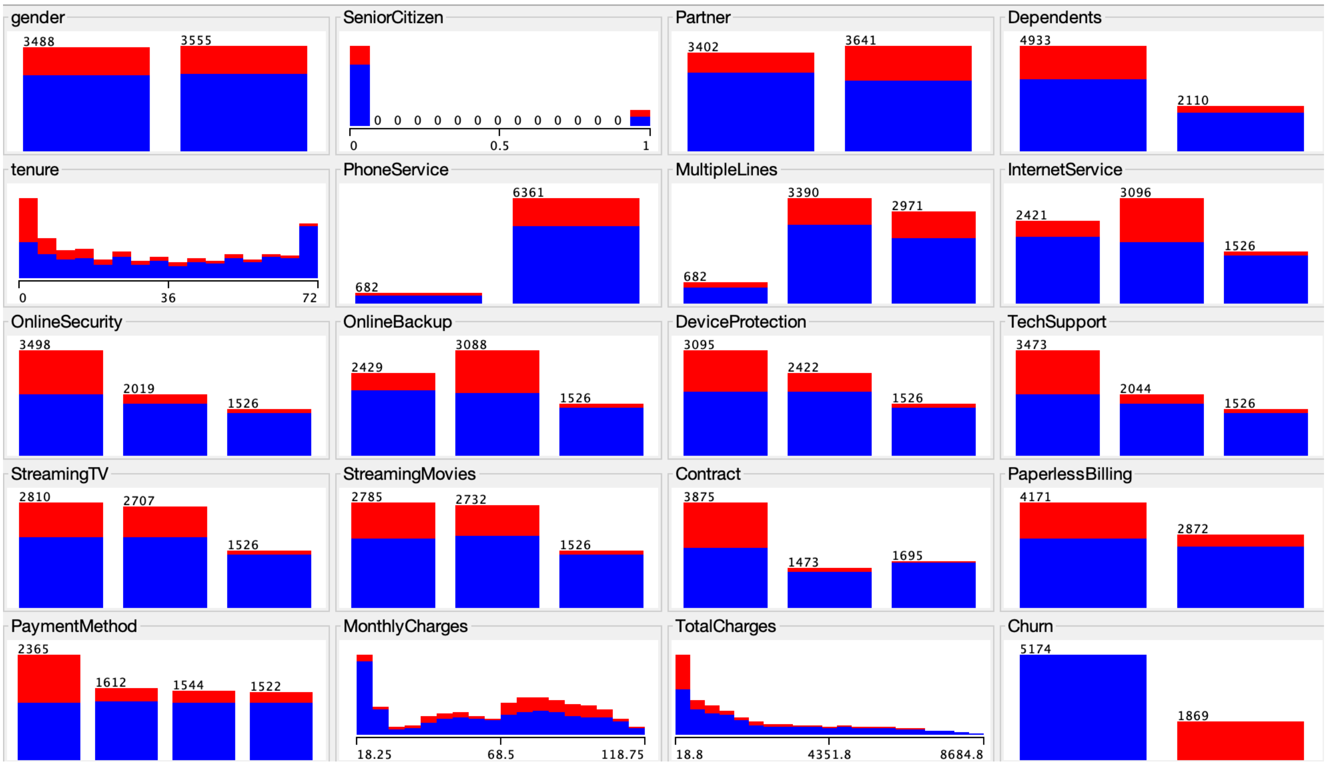
CUSTOMER CHURN PREDICTION: A BEHAVIORAL ANALYSIS



Summary

Customer churn presents a material risk to revenue stability in subscription-based businesses. This study evaluates the effectiveness of machine learning models in predicting churn using customer behavioral and account data.

Three models were developed and tested using cross-validation: Logistic Regression, Decision Tree (J48), and Random Forest.

While the leading model achieved c.80% accuracy, further analysis indicates that it fails to identify a substantial share of customers who ultimately churn. This exposes a critical misalignment between conventional model performance metrics and business value.

The findings demonstrate that model effectiveness in churn prediction is driven not by accuracy, but by the ability to identify at-risk customers early, with recall emerging as the most relevant metric. This aligns with existing research highlighting the limitations of accuracy in imbalanced classification problems (IEEE, 2024; Dimension Labs, n.d.).

Problem Statement

Customer churn occurs when a customer stops using a company's product or service. For telecom companies, churn can lead to:

- Revenue loss
- Increased acquisition costs
- Reduced long-term customer value

The objective of this project was to build a machine learning model capable of predicting whether a customer would churn based on behavioral and account-related data.

Previous studies emphasise that effective churn prediction enables proactive retention strategies and can significantly impact business performance (IEEE, 2023; Avaus, n.d.).

Dataset Overview

The dataset contains customer information from a telecom company and was sourced from a publicly available telecommunications churn dataset on Kaggle (BlastChar, n.d.).

Dataset characteristics
- Total instances: 7,043
- Target variable: Churn
- Type of problem: Binary classification

Key features
- Tenure
- MonthlyCharges
- TotalCharges
- Contract type
- Internet service

- Payment method
- Tech support
- Senior citizen status

Target classes

- No → customer stays
- Yes → customer leaves

The dataset was moderately imbalanced, with fewer customers in the churn category, a common characteristic in churn prediction problems that can affect model evaluation (JISEM Journal, 2023).

Dataset Preparation

Before training the models, several preprocessing steps were applied.

Preprocessing steps

- Removed irrelevant identifier field:
    - customerID
- Checked attribute types
- Converted target variable:
    - `Churn` set as class attribute
- Verified missing values
- Applied 10-fold cross-validation for evaluation

This ensured the dataset was suitable for model training and fair performance comparison. Cross-validation is widely used to improve model generalisability and reduce overfitting (IEEE, 2024).

Methodology

Three machine learning models were tested.

Model 1 — Logistic Regression

Used as a baseline model because of:

- simplicity
- interpretability

- strong performance in binary classification

Model 2 — J48 Decision Tree
Used to:

- Identify churn patterns
- generate interpretable rules

Model 3 — Random Forest
Used to:

- improve predictive performance
- reduce overfitting risk

These models are commonly applied in churn prediction due to their balance between interpretability and predictive capability (Kumo AI, n.d.; JISEM Journal, 2023).

Model Performance

Model	Accuracy	Churn	AUC
Logistic Regression	80.2%	0,545	0,845
Decision Tree	78%	0,490	0,758
Random Forest	78.7%	0,489	0,822

Logistic Regression achieved the best overall performance across all evaluation metrics, outperforming more complex models such as Random Forest.

This suggests that, for this dataset, simpler models can provide both strong predictive capability and greater interpretability, a finding consistent with prior research in structured datasets (IEEE, 2023).

Key Findings

The most important insight from this project was that:
Accuracy alone was misleading.

Although the Logistic Regression model achieved over 80% accuracy, it correctly identified only around 54% of customers who actually churned.

This means many customers at risk of leaving were still missed.

Why this matters

For churn prediction:

- Missing a churner can be costly
- Losing a customer may reduce long-term revenue
- Business value depends on detecting risk early

This shows why recall can be more important than accuracy in churn prediction.
This limitation is well documented in the literature, particularly in imbalanced datasets where models tend to favour the majority class (IEEE, 2024; Dimension Labs, n.d.).Business Interpretation

Business Interpretation

The analysis suggests that customer behavior can provide early warning signals of churn.
Potential churn indicators include:

- Short customer tenure
- Month-to-month contracts
- Higher monthly charges
- Lack of technical support
- Electronic payment methods

These findings can help businesses:

- improve retention strategies
- identify at-risk customers
- reduce customer loss

Industry research highlights that early identification of high-risk customers is critical for effective churn management (Pecan AI, n.d.; Avaus, n.d.).

Limitations

Several limitations were identified:

- The dataset was moderately imbalanced
- Some churn cases remained difficult to detect
- Accuracy did not fully represent business usefulness

Future improvements could include:

- class balancing techniques
- cost-sensitive learning
- feature engineering

Conclusion

This project set out to evaluate how effectively machine learning models can be used to predict customer churn, using behavioural and account-level data.

While the models achieved solid overall performance, the deeper analysis highlighted an important limitation: strong accuracy does not necessarily translate into practical usefulness. In this case, a significant proportion of churners remained undetected, limiting the model's ability to support real retention decisions.

This reinforces a key point in applied machine learning — performance should always be evaluated in the context of the problem being solved. For churn prediction, the priority is not simply to classify correctly overall, but to reliably identify customers who are at risk of leaving.

The results also showed that a relatively simple model, Logistic Regression, outperformed more complex alternatives. This suggests that model selection should not be driven purely by complexity, but by how well the model aligns with the structure of the data and the objective of the analysis.

From a business perspective, the implications are clear. Improving churn detection, even incrementally, can enable earlier and more targeted interventions, reducing customer loss and improving long-term value. However, achieving this requires a shift in focus — from maximising accuracy to optimising metrics that reflect real-world impact, particularly recall. These findings are consistent with existing research, which emphasises aligning model evaluation with business objectives rather than relying solely on statistical performance metrics (IEEE, 2023; Dimension Labs, n.d.).

Overall, this project demonstrates that the value of machine learning lies not only in building models, but in understanding their limitations, interpreting their outputs critically, and aligning their evaluation with business priorities.

References

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Appendix – Model Evaluation Outputs

A1. Logistic Regression — Evaluation Output

```
=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances        5650               80.2215 %
Incorrectly Classified Instances      1393               19.7785 %
Kappa statistic                          0.4644
Mean absolute error                      0.2714
Root mean squared error                  0.3683
Relative absolute error                 69.5915 %
Root relative squared error             83.4124 %
Total Number of Instances             7043

=== Detailed Accuracy By Class ===

                 TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
                 0,895    0,455    0,845      0,895    0,869      0,468    0,845     0,936     No
                 0,545    0,105    0,653      0,545    0,594      0,468    0,845     0,652     Yes
Weighted Avg.    0,802    0,362    0,794      0,802    0,796      0,468    0,845     0,861

=== Confusion Matrix ===

    a    b   <-- classified as
 4632  542 |    a = No
  851 1018 |    b = Yes
```

While the model performs strongly in identifying non-churn customers, its recall for churn cases remains limited (54.5%).

This indicates that a significant number of customers who eventually churn are not being detected, reinforcing the need to prioritise recall in this context.

A2. Decision Tree (J48) — Evaluation Output

```
=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances       5493              77.9923 %
Incorrectly Classified Instances     1550              22.0077 %
Kappa statistic                         0.3988
Mean absolute error                     0.2755
Root mean squared error                 0.4128
Relative absolute error                70.661  %
Root relative squared error            93.5016 %
Total Number of Instances            7043

=== Detailed Accuracy By Class ===

                 TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
                 0,885    0,510    0,828      0,885    0,855      0,403    0,758     0,849     No
                 0,490    0,115    0,606      0,490    0,541      0,403    0,758     0,516     Yes
Weighted Avg.    0,780    0,406    0,769      0,780    0,772      0,403    0,758     0,761

=== Confusion Matrix ===

    a    b   <-- classified as
 4578  596 |    a = No
  954  915 |    b = Yes
```

The Decision Tree model demonstrates reasonable performance in identifying non-churn customers; however, its ability to detect churn cases remains limited, with a recall of 49.0% for the churn class.

The confusion matrix highlights a high number of false negatives (954), indicating that a substantial proportion of customers who churn are not being identified.

While the model provides valuable interpretability through explicit decision rules, this comes at the cost of reduced predictive performance compared to Logistic Regression.

A3. Random Forest — Evaluation Output

```
=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances        5540               78.6597 %
Incorrectly Classified Instances      1503               21.3403 %
Kappa statistic                          0.4118
Mean absolute error                      0.2716
Root mean squared error                  0.3811
Relative absolute error                 69.647  %
Root relative squared error             86.3131 %
Total Number of Instances             7043

=== Detailed Accuracy By Class ===

                 TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
                 0,894    0,511    0,829      0,894    0,860      0,417    0,822     0,924     No
                 0,489    0,106    0,625      0,489    0,549      0,417    0,822     0,620     Yes
Weighted Avg.    0,787    0,403    0,775      0,787    0,778      0,417    0,822     0,843

=== Confusion Matrix ===

    a    b   <-- classified as
 4626  548 |    a = No
  955  914 |    b = Yes
```

The Random Forest model achieves a moderate level of overall performance, with an accuracy of 78.66% and a ROC AUC of 0.822.

However, similar to the Decision Tree model, its ability to detect churn cases is limited, with recall for the churn class at 48.9%.

The confusion matrix highlights a high number of false negatives (955), indicating that a substantial proportion of customers who churn are not identified.

Despite the use of an ensemble approach, the model does not demonstrate a meaningful improvement in churn detection compared to simpler methods.